

# Ten Edge-Degree Counts Are All a Bond-Incident-Degree Index Can See: An Exact Redundancy Certificate and What It Leaves for New Molecular Descriptors

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## Abstract

On molecular graphs an edge has one of ten degree types, so every bond-incident-degree index  $\sum_{uv \in E} f(d_u, d_v)$  — Zagreb, Randić, Sombor and their many variants — is a linear combination of the ten edge-type counts. We turn this classical fact into a *span certificate*: a rank condition on a baseline of classical indices under which every such index, whatever its formula or parameters, is an exact linear combination of the baseline and cannot add linear information about any target. The certificate holds for a standard baseline on all molecular graphs without hypervalent atoms; we verify it across the MoleculeNet benchmarks, where the single failure is diagnosed and recovered, and on a census of published degree-based indices, which also exposes a numerical pitfall that turns exact zeros into spurious partial correlations. For indices outside this class we give a target-aware screen with confidence intervals, a calibrated non-linear check and a chemistry-aware baseline, and find little residual signal. Finally, we show that hydrogen-suppressed graphs collide so often that every topology-only descriptor is bounded below what chemistry-aware descriptors already reach.

**Keywords.** Topological index; degree-based index; Sombor index; redundancy; partial correlation; QSPR; molecular descriptors.

**Code and data.** <https://github.com/Singati2/topological-index-orthogonality>

## 1 Introduction

Scalar topological indices have grown from a handful of early proposals [1, 2, 3] to several hundred, and the proliferation has been criticised within mathematical chemistry itself [4]. The growth is fastest for degree-based indices: since the Sombor index was introduced in 2021 [5], modified,  $p$ -, KG-, elliptic, Euler, diminished and hyperbolic Sombor variants have followed [6, 7, 8, 9, 10, 11, 12]. A new index is commonly supported by a correlation with a property on a small isomer set, by bounds and extremal graphs, and by its correlations with existing indices, e.g. intercorrelation tables among Sombor-type variants [24, 9]. Quality criteria that go beyond single correlations exist — structure sensitivity and abruptness [13], applicability tests on octanes [14], and clustering-based usefulness scores [15, 16] — but none of them answers the question a QSPR modeller needs answered: *does the new index carry information about a target that an explicit baseline does not already carry?*

For the largest class of indices this question turns out to be settled before any data are examined. It is classical that a bond-incident-degree (BID) index  $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$  is the linear combination  $\sum_{i \leq j} f(i, j) m_{ij}(G)$  of the edge-degree-pair counts  $m_{ij}$  [4, 17], and

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the resulting finite-dimensional linear view of vertex-degree-based indices has recently been developed in general [18, 19], including the observation that the analytic form of  $f$  is irrelevant once its values on the degree pairs are fixed. If a baseline of classical degree-based indices spans the edge-degree-pair counts realised in a dataset — a rank condition that does not involve the candidate or the target — then *every* BID index, whatever its functional form or parameters, is an exact linear combination of the baseline on that dataset, and a target-aware redundancy screen has nothing to measure. We call this rank condition a *span certificate*: a baseline is *certified* on a dataset when the condition holds, and a candidate is then *certified redundant*.

### Contributions.

- (1) *Span certificate* (Section 4). The mathematics is not ours: the linearity of BID indices in the counts  $m_{ij}$  and the finite-dimensional framework built on it are due to Gutman [4], Rada and Cruz [17] and Rada [19], and that framework contains no statistical or dataset-level analysis. What is ours is the statistical reading: an explicit, checkable rank condition on a baseline under which every BID index has zero residual, infinite variance inflation and no residual direction towards any target (Proposition 4.2); its data-free form for graphs of maximum degree at most four, with the explicit coefficient vectors (Corollary 4.3, which we have not found stated in this form); the observation that a standard 18-index baseline satisfies it; and its verification on benchmark data.
- (2) *Generality* (Section 4.1). The certificate is checked on 13 MoleculeNet SMILES datasets (318 648 molecules): it holds on 12, and the one failure (HIV) is caused by 350 organometallic molecules with atoms of degree five to ten — the case the data-free certificate excludes and the data-level check catches.
- (3) *Census* (Section 5). Fourteen published degree-based indices (15 instances, eight of them Sombor-type) and 100 points of a three-parameter family, on four MoleculeNet datasets: all exactly redundant, which sharpens earlier correlational reports of near-redundancy into an exact statement; two non-BID Sombor-type controls correctly fall outside the span.
- (4) *A numerical pitfall and a tool* (Section 6). For exactly redundant indices, partial correlations computed with absolute residual tolerances are noise of order  $10^{-2}$ ; we give a numerically safe recipe and a one-function implementation of the certificate that returns each index’s coefficient vector.
- (5) *Target-aware screen for non-BID indices* (Section 7). Confidence intervals against an effect-size margin, a cross-fitted non-linear residualization with a proper null distribution, and a repeat of the screen against a chemistry-aware (RDKit 2D) baseline, on 20 alternative candidates.
- (6) *Information ceiling* (Section 8). Graph collisions under hydrogen suppression, confirmed by exact isomorphism, give an in-sample oracle for the whole topology-only representation; a graph-grouped, corrected benchmark places the indices against this oracle and against an atom- and bond-aware descriptor reference.

One of the parametric families screened below, the three-parameter Loyola family of Section 5, is studied mathematically in separate work by a subset of the present authors; here it serves only as a generic parametric BID family, and the span certificate explains why no such family can add linearly to a classical baseline.

## 2 Related work

**Redundancy and orthogonalization.** Strong mutual correlation among topological indices is long established. Hollas studied the redundancy of graph invariants used in chemoinformatics [20] and related the covariance of degree-based indices to the edge structure of the graph [21];

orthogonalization of descriptors is an equally old remedy [22, 23]. Recent work continues to report intercorrelations among new variants, e.g. among Sombor-type indices on octanes [9, 24], and large-scale descriptor evaluations cluster descriptors without reference to a target [15, 16]. We do not claim to discover that topological indices are redundant.

**Linear structure of degree-based indices.** The identity  $I_f = \sum f(i, j)m_{ij}$  is standard [4, 17] and underlies the bond-additive modelling programme in which the term “bond incident degree” index originates [25, 26]. Matrix and finite-dimensional linear formulations of vertex-degree-based indices, including their discrimination properties, are developed in [18, 19]; the M-polynomial [27] is the generating function of the same counts; further algebraic constructions on degree-based indices appear in [28]; and the edge-type-count view underlies recent extremal-graph work for chemical graphs of maximum degree three [31, 29, 30]. The boundary between that literature and the present paper is stated once, in Contribution (1).

**Degeneracy.** That distinct graphs can share index values (degeneracy) and that hydrogen suppression merges distinct molecules are classical concerns [33, 34, 35]; a recent audit of the MoleculeNet benchmarks documents duplicate and conflicting entries at the SMILES level [32]. Section 8 quantifies collisions at the level of the hydrogen-suppressed graph (isomorphism classes), which is the relevant level for topological indices, and uses them as a ceiling for the whole topology-only representation rather than for a single index.

**Descriptor evaluation in QSPR.** Topological descriptors remain competitive baselines in modern property prediction [36, 37], which keeps the question of which indices are worth adding practically relevant. Generic feature selection (LASSO [38], mRMR [39], Boruta [40]) acts on a fixed descriptor matrix; the screen here acts one layer earlier, on whether a proposed index should enter that matrix at all.

### 3 Data, curation and baseline

We use four MoleculeNet benchmarks [41]: ESOL (aqueous solubility) [42], FreeSolv (experimental hydration free energy; column `expt`) [43], Lipophilicity ( $\log D_{7.4}$ ) [44] and BBBP (blood–brain barrier penetration, binary) [45]. SMILES are parsed with RDKit [46]; the largest fragment is kept, the hydrogen-suppressed, atom- and bond-unlabelled graph is formed, and molecules with fewer than two heavy atoms are dropped, so that every graph is connected and has no isolated vertex. On the files as distributed this leaves 1127, 639, 4200 and 2039 molecules (the “parsed” column of Table 1). Table 1 reports the counts. Stereochemistry and atom identity are invisible to every descriptor studied here, so molecules that differ only in these respects become duplicates; we report all screening results on the files as distributed and verify in Section 7 that de-duplication on canonical SMILES (averaging regression duplicates whose labels agree within 0.5 target units, dropping duplicate groups that disagree by more or carry conflicting BBBP labels) changes no conclusion; the benchmark of Section 8 uses the curated data. All molecules have maximum degree at most four after hydrogen suppression.

**Baseline.** The 30-index baseline (Supplementary Table S1) contains 18 degree-based indices, 8 distance-based indices, 3 spectral indices and the Balaban index. All 18 degree-based indices are BID indices in the sense of Section 4 (vertex-degree sums  $\sum_v g(d_v)$  are BID with  $f(a, b) = g(a)/a + g(b)/b$  on graphs without isolated vertices). Several baseline members are related by exact identities, which we disclose because they matter for numerical rank:  $\sigma = F - 2M_2$  and  $\text{red}M_1 = M_1 - 4|E(G)| + |V(G)|$ . The index labelled “Schultz” in our code is the degree distance  $\sum_{\{u,v\}} (d_u + d_v) d(u, v)$  (the Schultz molecular topological index differs by  $M_1$ ); PI is the edge Padmakar–Ivan index; the augmented Zagreb summand is set to zero on  $K_2$  edges.

Table 1: Datasets and curation. Raw rows of the MoleculeNet files; invalid SMILES; multi-fragment SMILES reduced to the largest fragment; molecules parsed to a graph with at least two heavy atoms (the sets used by the screens); rows in exact canonical-SMILES duplicate groups; duplicate groups dropped for conflicting labels; molecules after curation (also dropping graphs with fewer than two heavy atoms); distinct hydrogen-suppressed unlabelled graphs (Weisfeiler–Lehman hash); and the percentage of curated molecules whose graph is shared with another molecule.

| Dataset       | Target                             | raw  | invalid | multi-frag. | parsed | dup. rows | conflicts | curated | graphs | shared (%) |
|---------------|------------------------------------|------|---------|-------------|--------|-----------|-----------|---------|--------|------------|
| ESOL          | $\log S$ (mol/L)                   | 1128 | 0       | 0           | 1127   | 22        | 1         | 1115    | 628    | 55.5       |
| FreeSolv      | $\Delta G_{\text{hyd}}$ (kcal/mol) | 642  | 0       | 0           | 639    | 0         | 0         | 639     | 236    | 74.3       |
| Lipophilicity | $\log D_{7.4}$                     | 4200 | 0       | 1           | 4200   | 0         | 0         | 4200    | 3472   | 28.2       |
| BBBP          | BBB penetration (0/1)              | 2050 | 11      | 105         | 2039   | 141       | 11        | 1955    | 1693   | 22.6       |

The composition of the baseline is listed in Supplementary Table S1.

## 4 The span certificate

Let  $\mathcal{G}_\Delta$  be the class of finite simple graphs without isolated vertices and with maximum degree at most  $\Delta$ , and  $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$ , so  $|P_\Delta| = 10$ . For  $G \in \mathcal{G}_\Delta$  let  $m_{ij}(G)$  be the number of edges whose end degrees are  $\{i, j\}$ . A *BID index* is  $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$  for a symmetric  $f : \{1, \dots, \Delta\}^2 \rightarrow \mathbb{R}$ .

**Remark 4.1** (classical).  $I_f(G) = \sum_{(i,j) \in P_\Delta} f(i, j) m_{ij}(G)$ , because the edge classes  $\{uv : \{d_u, d_v\} = \{i, j\}\}$  partition  $E(G)$  and  $f$  is constant on each [4, 17]. For example, the path on four vertices has two edges of type  $(1, 2)$  and one of type  $(2, 2)$ , so  $I_f = 2f(1, 2) + f(2, 2)$  on it for every  $f$ , and the star  $K_{1,3}$  has three edges of type  $(1, 3)$ , so  $I_f(K_{1,3}) = 3f(1, 3)$ . On  $\mathcal{G}_4$  the ten functionals  $m_{ij}$  are linearly independent: the complete bipartite graph  $K_{i,j}$  has all  $ij$  edges of type  $(i, j)$ , so its count vector is a multiple of the  $(i, j)$ -th unit vector. Hence the BID indices on  $\mathcal{G}_4$  form exactly a ten-dimensional space; on connected graphs with at least three vertices  $m_{11} = 0$  and the space is at most nine-dimensional.

Now fix a dataset  $D = \{G_1, \dots, G_n\} \subset \mathcal{G}_\Delta$  with realised pair set  $P_D \subseteq P_\Delta$ , count matrix  $M \in \mathbb{R}^{n \times |P_D|}$ , and a baseline whose BID block consists of  $k$  indices with value matrix  $\Phi_B \in \mathbb{R}^{k \times |P_D|}$ ,  $(\Phi_B)_{b,(i,j)} = f_b(i, j)$ . The baseline’s BID columns are  $B = M\Phi_B^\top \in \mathbb{R}^{n \times k}$ . Write  $\mathbf{1}$  for the all-ones vector.

The following is an elementary consequence of Remark 4.1; its statistical reading (i)–(iv) is the point.

**Proposition 4.2** (Span certificate). *If  $\text{rank}[\mathbf{1}, B] = \text{rank}[\mathbf{1}, M]$ , then for every symmetric  $f$  the vector  $z_f = (I_f(G_t))_{t \leq n}$  lies in  $\text{col}[\mathbf{1}, B]$ . Consequently, for any baseline matrix  $X$  whose columns include  $B$  and for every target vector  $y$ : (i) the least-squares residual of  $z_f$  on  $[\mathbf{1}, X]$  is zero; (ii) if  $z_f$  is not constant on  $D$ , its variance inflation relative to  $X$  is infinite; (iii) the partial correlation  $\text{pcor}(z_f, y \mid X)$  — the correlation between the residuals of  $z_f$  and of  $y$  after least-squares regression on  $[\mathbf{1}, X]$  — is undefined (0/0):  $z_f$  carries no direction beyond  $X$ , and we report it as 0 by convention — any nonzero value computed in floating point is an artefact (Section 6); and (iv) adding  $z_f$  to  $X$  leaves every ordinary least-squares fit (fitted values, residuals,  $R^2$ ) on  $D$  unchanged. Penalised linear fits (ridge, LASSO) can change, but only through the geometry of the penalty, not through any new direction in the data.*

*Proof.* By Remark 4.1,  $z_f = Mf_D$  with  $f_D = (f(i, j))_{(i,j) \in P_D}$ , so  $z_f \in \text{col}(M)$ . Since  $B = M\Phi_B^\top$ ,  $\text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, M]$ , and equal ranks give equality. Thus  $z_f \in \text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, X]$ , which is (i); (ii) and (iii) restate (i), and (iv) holds because an OLS fit depends on the design only through its column space.  $\square$

**Corollary 4.3** (data-free certificate on  $\mathcal{G}_4$ ). *If  $\text{rank } \Phi_B = 10$  over  $P_4$ , then for every symmetric  $f$  there is a coefficient vector  $c$ , unique up to  $\ker \Phi_B^\top$ , with  $\Phi_B^\top c = f_{P_4}$ , and  $I_f(G) = \sum_b c_b I_{f_b}(G)$  holds identically on  $\mathcal{G}_4$ . In particular the conclusions of Proposition 4.2 hold on every dataset of graphs in  $\mathcal{G}_4$ , without inspecting the data.*

*Proof.*  $\text{rank } \Phi_B = 10$  means the rows  $f_b$  span  $\mathbb{R}^{P_4}$ ; then

$$I_f = \sum_{ij} f(i, j) m_{ij} = \sum_{ij} \sum_b c_b f_b(i, j) m_{ij} = \sum_b c_b I_{f_b}.$$

□

**The baseline satisfies the certificate.** Evaluating each of the 18 degree-based baseline indices on  $K_{i,j}$  gives  $\Phi_B$  exactly (Supplementary Table S2); its rank over  $P_4$  is 10 (smallest-to-largest singular value ratio  $1.2 \times 10^{-4}$ ), so Corollary 4.3 applies. The certificate is generic rather than delicate: 37 322 of the 43 758 ten-element subsets of the 18 indices already have rank 10, for example  $\{M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA, AG\}$ . On the data, the numerical ranks of  $M$ ,  $B$  and  $[B, M]$  coincide on all four datasets (Table 2).

**Remark 4.4** (what the certificate does and does not say). The certificate is a statement about *linear* redundancy. It does not say that a BID index is useless: a parsimonious index can summarise the ten counts well, and a model that is not rotation-invariant (e.g. a tree ensemble) may use a BID index differently from its constituents. It does say that no BID index can add linearly to a certified baseline, so linear redundancy diagnostics — pairwise correlation, the variance inflation factor (VIF), partial correlation, and the gain of an ordinary least-squares model — are decided in advance for this class. The certificate covers edge-additive indices only; it does not cover normalised or averaged indices such as  $I_f/|E|$ , multiplicative indices  $\prod f$ , non-linear transforms such as  $\log I_f$ , products of indices, neighbourhood-degree-sum and other second-order indices, or valence- and heteroatom-weighted degrees. Beyond  $\Delta = 4$  (hypervalent atoms, explicit hydrogens, coordination compounds) the space has dimension  $\Delta(\Delta + 1)/2$  and the certificate must be checked on the data, which Proposition 4.2 makes a one-line rank computation.

Table 2: Span certificate on the four datasets.  $|P_D|$ : realised degree pairs; ranks are numerical ranks of the column-centred, column-normalised matrices (relative tolerance  $10^{-9}$ ). “Resid.” columns give the largest relative residual  $\|r\|/\|z - \bar{z}\|$  on the baseline of the ten counts  $m_{ij}$ , of the 15 published BID instances of Table 4, and of the 100 Loyola grid points.

| Dataset       | $n$  | $ P_D $ | rk $M$ | rk $B$ | rk $[B, M]$ | resid. $m_{ij}$       | resid. published      | resid. Loyola         |
|---------------|------|---------|--------|--------|-------------|-----------------------|-----------------------|-----------------------|
| ESOL          | 1127 | 10      | 10     | 10     | 10          | $4.5 \times 10^{-13}$ | $1.9 \times 10^{-13}$ | $6.9 \times 10^{-13}$ |
| FreeSolv      | 639  | 10      | 10     | 10     | 10          | $2.1 \times 10^{-12}$ | $5.6 \times 10^{-14}$ | $1.3 \times 10^{-13}$ |
| Lipophilicity | 4200 | 9       | 9      | 9      | 9           | $2.6 \times 10^{-12}$ | $3.5 \times 10^{-13}$ | $1.9 \times 10^{-12}$ |
| BBBP          | 2039 | 10      | 10     | 10     | 10          | $1.1 \times 10^{-12}$ | $1.8 \times 10^{-13}$ | $1.3 \times 10^{-12}$ |

#### 4.1 Scope check across MoleculeNet

The data-free certificate covers every dataset of graphs with maximum degree four. To see how far that reaches, and what happens beyond it, we ran the data-level check of Proposition 4.2 on every SMILES-based MoleculeNet dataset we could obtain (Table 3; only SMILES are used, no targets; 318 648 molecules in all). For the eight datasets whose molecules all have maximum degree four — including the 133 882 molecules of QM9 and the 93 087 of MUV — Corollary 4.3 guarantees the outcome, and the run is a numerical check of it: the ranks of  $M$  and  $B$  agree and

every  $m_{ij}$  column has relative residual at most  $1.6 \times 10^{-9}$ . The informative cases are the other five. Four datasets (ClinTox, SIDER, Tox21, ToxCast) contain one to three molecules with an atom of degree five or six; here certification is data-dependent, because as functions on  $P_6$  the 18 baseline functions have rank only 17 ( $\sigma = F - 2M_2$  exactly), so at most seven pairs beyond  $P_4$  can ever be absorbed. These datasets realise one to three such pairs, the baseline’s rank rises by the same amount, and the certificate holds. The single failure is HIV: 350 of its 41 127 molecules are organometallic complexes (Co, Ni, Cu, Rh, Fe, Mn, Cr, Mo, Ir centres with five to ten bonds, and pentavalent phosphorus), which realise 37 degree pairs; the count matrix has rank 36 while the 18-index baseline has rank 17, so no baseline of 18 BID indices can certify this dataset. On the 40 777 HIV molecules of maximum degree four the certificate holds again (ranks 10/10, residual  $5.0 \times 10^{-12}$ ). This is the intended behaviour: the data-free certificate is a statement about  $\mathcal{G}_4$ , and the one-line data-level rank check is what detects a dataset that leaves that class.

Table 3: The span certificate on thirteen MoleculeNet SMILES datasets (no targets used).  $n$ : parsed molecules;  $n_{>4}$ : molecules containing an atom of degree above four;  $\Delta$ : largest degree;  $|P_D|$ : realised degree pairs; ranks of the count matrix  $M$  and of the 18-index baseline block  $B$ ; largest relative residual of an  $m_{ij}$  column on  $B$ ; whether  $\text{rank}[\mathbf{1}, B] = \text{rank}[\mathbf{1}, M]$  with residuals below  $10^{-8}$ .

| Dataset       | $n$    | $n_{>4}$ | $\Delta$ | $ P_D $ | rk $M$ | rk $B$ | max resid.            | certified                                                                 |
|---------------|--------|----------|----------|---------|--------|--------|-----------------------|---------------------------------------------------------------------------|
| ESOL          | 1127   | 0        | 4        | 10      | 10     | 10     | $4.6 \times 10^{-13}$ | yes                                                                       |
| FreeSolv      | 639    | 0        | 4        | 10      | 10     | 10     | $2.1 \times 10^{-12}$ | yes                                                                       |
| Lipophilicity | 4200   | 0        | 4        | 9       | 9      | 9      | $2.6 \times 10^{-12}$ | yes                                                                       |
| BBBP          | 2039   | 0        | 4        | 10      | 10     | 10     | $1.1 \times 10^{-12}$ | yes                                                                       |
| BACE          | 1513   | 0        | 4        | 9       | 9      | 9      | $1.8 \times 10^{-12}$ | yes                                                                       |
| ClinTox       | 1477   | 1        | 6        | 11      | 11     | 11     | $2.4 \times 10^{-12}$ | yes (on the $\Delta \leq 4$ subset, $n = 1476$ : ranks 10/10, yes)        |
| SIDER         | 1396   | 1        | 6        | 11      | 11     | 11     | $7.7 \times 10^{-12}$ | yes (on the $\Delta \leq 4$ subset, $n = 1395$ : ranks 10/10, yes)        |
| Tox21         | 7812   | 2        | 6        | 12      | 12     | 12     | $6.6 \times 10^{-11}$ | yes (on the $\Delta \leq 4$ subset, $n = 7810$ : ranks 10/10, yes)        |
| ToxCast       | 8566   | 3        | 6        | 13      | 13     | 13     | $1.6 \times 10^{-9}$  | yes (on the $\Delta \leq 4$ subset, $n = 8563$ : ranks 10/10, yes)        |
| QM8           | 21783  | 0        | 4        | 10      | 10     | 10     | $3.6 \times 10^{-11}$ | yes                                                                       |
| QM9           | 133882 | 0        | 4        | 10      | 10     | 10     | $1.4 \times 10^{-10}$ | yes                                                                       |
| MUV           | 93087  | 0        | 4        | 9       | 9      | 9      | $2.0 \times 10^{-11}$ | yes                                                                       |
| HIV           | 41127  | 350      | 10       | 37      | 36     | 17     | $9.9 \times 10^{-1}$  | <b>no</b> (on the $\Delta \leq 4$ subset, $n = 40777$ : ranks 10/10, yes) |

**A one-function implementation.** The certificate and the coefficient vectors are available as a single function, `certify(molecules, f, baseline)` in `src/certify.py`, which returns the data-free rank of  $\Phi_B$ , the realised pairs, the data-level ranks and residuals, and, for a candidate  $f$ , its coefficient vector on the baseline with the identity error on the data. For the Euler Sombor index on five test molecules the identity error is  $7 \times 10^{-16}$ ; for the hyper-Zagreb index on the ten-index basis it returns  $HM = F + 2M_2$  exactly. The  $18 \times 10$  value matrix  $\Phi_B$  is listed in the Supplementary Information (Table S2).

## 5 Census of published degree-based indices

We screened the 15 instances of 14 published BID indices in Table 4 — eight Sombor-type instances (the  $p$ -Sombor index at two values of  $p$ ) and seven other degree-based indices — and all 100 points of the Loyola grid  $(\alpha, \beta, \gamma) \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}^2 \times \{0, \frac{1}{2}, 1, 2\}$  of  $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp(\gamma |d_u - d_v| / (d_u + d_v))$  (a three-parameter BID family that reduces to the Gutman–Milovanović family at  $\gamma = 0$ ) on each of the four datasets. As controls outside the BID class we added the Sombor coindex (sum over non-adjacent pairs) and the eccentric Sombor index (eccentricities in place of degrees).

Every published BID index lies in the span of the baseline to a relative residual of at most  $4 \times 10^{-13}$ , and every Loyola grid point to at most  $2 \times 10^{-12}$ ; the two controls have relative

Table 4: Census of published BID indices ( $f(a, b)$  with  $a, b$  the end degrees of an edge) and two non-BID Sombor-type controls. Columns 3–6: largest  $|r|$  with any of the 30 baseline indices on each dataset (the conventional pairwise screen). Last column: largest relative residual on the baseline over the four datasets. For KG-Sombor  $e = a + b - 2$  is the edge degree. Every BID index is in the span to machine precision; the controls are not.

| Index                               | $f(a, b)$                             | ESOL  | FreeSolv | Lipo  | BBBP  | max resid.            |
|-------------------------------------|---------------------------------------|-------|----------|-------|-------|-----------------------|
| Modified Sombor [6]                 | $1/\sqrt{a^2 + b^2}$                  | 0.999 | 0.999    | 0.999 | 1.000 | $8.6 \times 10^{-14}$ |
| Elliptic Sombor [7]                 | $(a + b)\sqrt{a^2 + b^2}$             | 1.000 | 0.999    | 0.999 | 1.000 | $4.1 \times 10^{-14}$ |
| Euler Sombor [8]                    | $\sqrt{a^2 + b^2 + ab}$               | 1.000 | 1.000    | 1.000 | 1.000 | $6.8 \times 10^{-14}$ |
| Diminished Sombor [9]               | $\sqrt{a^2 + b^2}/(a + b)$            | 1.000 | 1.000    | 1.000 | 1.000 | $4.6 \times 10^{-14}$ |
| Hyperbolic Sombor [10]              | $\sqrt{a^2 + b^2}/\min(a, b)$         | 0.987 | 0.983    | 0.985 | 0.990 | $3.5 \times 10^{-13}$ |
| $p$ -Sombor, $p = \frac{1}{2}$ [11] | $(a^{1/2} + b^{1/2})^2$               | 1.000 | 1.000    | 1.000 | 1.000 | $5.9 \times 10^{-14}$ |
| $p$ -Sombor, $p = 3$ [11]           | $(a^3 + b^3)^{1/3}$                   | 1.000 | 1.000    | 1.000 | 1.000 | $4.6 \times 10^{-14}$ |
| KG-Sombor [12]                      | $\sqrt{a^2 + e^2} + \sqrt{b^2 + e^2}$ | 0.999 | 0.999    | 0.999 | 0.999 | $1.1 \times 10^{-13}$ |
| Inverse sum indeg [25]              | $ab/(a + b)$                          | 0.998 | 0.997    | 0.998 | 0.998 | $1.2 \times 10^{-13}$ |
| Symmetric division deg [25]         | $a/b + b/a$                           | 0.993 | 0.992    | 0.992 | 0.994 | $1.1 \times 10^{-13}$ |
| Nirmala [54]                        | $\sqrt{a + b}$                        | 1.000 | 1.000    | 1.000 | 1.000 | $4.7 \times 10^{-14}$ |
| First inverse Nirmala [55]          | $\sqrt{1/a + 1/b}$                    | 0.999 | 0.999    | 0.999 | 0.999 | $4.9 \times 10^{-14}$ |
| Second inverse Nirmala [55]         | $\sqrt{ab/(a + b)}$                   | 1.000 | 0.999    | 0.999 | 1.000 | $1.8 \times 10^{-13}$ |
| First Gourava [56]                  | $a + b + ab$                          | 0.999 | 0.999    | 0.999 | 0.999 | $4.6 \times 10^{-14}$ |
| Hyper-Zagreb [57]                   | $(a + b)^2$                           | 0.999 | 0.998    | 0.998 | 0.999 | $4.1 \times 10^{-14}$ |
| Sombor coindex (control)            | non-BID                               | 0.999 | 0.998    | 0.997 | 0.999 | $7.7 \times 10^{-3}$  |
| Eccentric Sombor (control)          | non-BID                               | 0.971 | 0.976    | 0.943 | 0.947 | $7.5 \times 10^{-2}$  |

residuals of 0.006–0.008 (coindex) and 0.06–0.08 (eccentric Sombor), i.e. they are correctly *not* certified. The universal coefficients of Corollary 4.3 reproduce the data-level values of all 15 instances to relative error below  $10^{-10}$  (file `results/bid_universal_identities.csv`). Two of them reduce to familiar identities,  $GG_1 = M_1 + M_2$  and  $HM = F + 2M_2$ ; the others have non-sparse coefficient vectors, which is the generic situation and the reason the redundancy is invisible to a pairwise inspection.

The census is a numerical verification of Corollary 4.3, not independent evidence for it. The pairwise screen happens to reject all 15 published instances here (their maximal correlation with a baseline index is 0.98–1.00), and it passes only one of the 400 Loyola evaluations (100 grid points on four datasets) (FreeSolv,  $(\alpha, \beta, \gamma) = (-1, -1, 1)$ ,  $\max |r| = 0.943$ ). The certificate reaches the same verdicts for a different reason: it is exact, needs no threshold, and would apply equally to a variant whose pairwise correlations happened to fall below any threshold.

## 6 A numerical pitfall

An earlier draft of this work reported maximal partial correlations of 0.028 (ESOL), 0.036 (FreeSolv) and 0.009 (Lipophilicity) for the Loyola grid. By Proposition 4.2 the true value is zero. The nonzero values arose because a residual was declared zero only when its absolute standard deviation fell below  $10^{-12}$ ; for indices of magnitude  $10^2$ – $10^4$  the least-squares residual of an exactly redundant index is floating-point noise of relative size  $10^{-15}$ – $10^{-12}$  but absolute size above that tolerance, and correlating this noise with the target residual gives values of order  $1/\sqrt{n}$ . Partial correlations are the most affected, because they are a ratio of two noise quantities; VIF and incremental  $R^2$  computed in the same way appear as finite but meaningless numbers (very large, respectively very small). We recommend (a) a relative criterion  $\|r_z\|/\|z - \bar{z}\| < 10^{-8}$  with a rank-revealing solver, (b) reporting the in-span status explicitly, and (c) for BID candidates, replacing the statistical computation by the certificate. Our implementation (`src/orthogonality.py`, `span_certified_pcor`) does all three; with it, all 400 Loyola evaluations are exactly 0.

## 7 A target-aware screen for indices outside the BID class

For a candidate  $z$  that is not certified, the question is whether it carries information about a target  $y$  beyond the baseline  $X$ . We report three quantities.

- (a) *Pairwise screen*:  $\max_b |r(z, X_b)| < 0.95$ , kept only for comparability with current practice.
- (b) *Linear partial correlation with a confidence interval*:  $\text{pcor}(z, y \mid X)$  from OLS residuals on  $[\mathbf{1}, X]$ , with a Fisher- $z$  interval with standard error  $1/\sqrt{n - \text{rk } X - 3}$ , where  $\text{rk } X$  is the numerical rank of  $X$ , read against an effect-size margin  $\tau = 0.10$  fixed before the analysis: *pass* if the 95% interval for  $|\text{pcor}|$  lies above  $\tau$ , *negligible* if it lies below, *inconclusive* otherwise. The margin is a convention for a “one percent of residual variance” effect, not a significance threshold.
- (c) *Non-linear conditional association*:  $z$  and  $y$  are each residualized on  $X$  by cross-fitted random forests (five folds), and the correlation  $\rho_{\text{nl}}$  of the two residual vectors is reported with its Fisher- $z$  interval, in the spirit of double machine learning and the generalised covariance measure [49, 50]. Because the forests are only approximate, we calibrate against three references computed in the same way: an exactly redundant BID index, a pure-noise column, and a deliberately leaky column  $y + \varepsilon$ .

The 20 candidates span information-theoretic, centrality, spectral-hybrid, motif and eccentricity families (Supplementary Information, Section S3); none is claimed as new, and none is a BID index. (Harmonic centrality is identically twice the Harary index and is correctly flagged as in span.)

Table 5: Target-aware screen of the 20 non-BID candidates. Top: counts per dataset — pairwise pass ( $\max |r| < 0.95$ ), point-estimate combined pass ( $|\text{pcor}| \geq 0.10$  and pairwise pass), interval pass (pairwise pass and 95% interval of  $|\text{pcor}|$  above 0.10), inconclusive, negligible, and in span. Bottom: every candidate–dataset pair with  $|\text{pcor}| \geq 0.10$  or  $|\rho_{\text{nl}}| \geq 0.10$ , with 95% Fisher- $z$  intervals.

| Dataset       |  | pairwise | point comb. | interval pass | inconcl. | neglig. | in span |
|---------------|--|----------|-------------|---------------|----------|---------|---------|
| ESOL          |  | 15       | 4           | 0             | 12       | 7       | 1       |
| FreeSolv      |  | 15       | 1           | 0             | 15       | 4       | 1       |
| Lipophilicity |  | 16       | 0           | 0             | 1        | 18      | 1       |
| BBBP          |  | 17       | 2           | 2             | 1        | 16      | 1       |

| Dataset  | Candidate  | $\max  r $ | linear pcor [95% CI]    | verdict      | $\rho_{\text{nl}}$ [95% CI] |
|----------|------------|------------|-------------------------|--------------|-----------------------------|
| ESOL     | Bonchev_Id | 0.992      | −0.113 [−0.171, −0.054] | inconclusive | +0.015 [−0.043, +0.074]     |
| ESOL     | CutVerts   | 0.833      | −0.118 [−0.176, −0.059] | inconclusive | −0.015 [−0.074, +0.043]     |
| ESOL     | FourCyc    | 0.189      | +0.112 [+0.054, +0.170] | inconclusive | +0.001 [−0.058, +0.059]     |
| ESOL     | InfoH_deg  | 0.699      | +0.154 [+0.096, +0.211] | inconclusive | +0.142 [+0.085, +0.199]     |
| ESOL     | MeanEcc    | 0.930      | −0.109 [−0.167, −0.050] | inconclusive | −0.028 [−0.087, +0.030]     |
| FreeSolv | EE_x_W     | 0.985      | +0.129 [+0.051, +0.206] | inconclusive | +0.027 [−0.050, +0.105]     |
| FreeSolv | InfoH_deg  | 0.649      | −0.061 [−0.140, +0.018] | inconclusive | −0.104 [−0.181, −0.027]     |
| FreeSolv | Triangles  | 0.126      | +0.107 [+0.028, +0.184] | inconclusive | +0.011 [−0.067, +0.089]     |
| BBBP     | FourCyc    | 0.224      | −0.220 [−0.261, −0.178] | pass         | −0.171 [−0.212, −0.128]     |
| BBBP     | Triangles  | 0.119      | +0.225 [+0.183, +0.266] | pass         | +0.013 [−0.031, +0.056]     |

Calibration of  $\rho_{\text{nl}}$  across the four datasets: exactly redundant BID reference ( $\sum (d_u + d_v)^{-1/2}$ , i.e. SCI) −0.044 to +0.043; pure noise −0.030 to +0.039; leaky column  $y + \varepsilon$  0.567 to 0.661.

**Results.** The pairwise screen admits 15, 15, 16 and 17 of the 20 candidates on ESOL, FreeSolv, Lipophilicity and BBBP. Point estimates  $|\text{pcor}| \geq 0.10$  would add four ESOL candidates and one FreeSolv candidate, but with intervals none of them clears the margin: they are inconclusive,

with intervals straddling 0.10 (Table 5). This is largely a matter of power: the half-width of a 95% interval is about 0.08 at  $n = 639$  (FreeSolv) and 0.06 at  $n = 1127$  (ESOL), so a pass would require  $|\text{pcor}|$  of about 0.18 and 0.16. On Lipophilicity ( $n = 4200$ ) 18 of the 20 are negligible. Only on BBBP do two candidates, the triangle and four-cycle counts, clear the margin with their whole interval (linear-probability partial correlations 0.225 and  $-0.220$ ).

The non-linear check separates these further. The calibration references behave as intended: the leaky column gives  $\rho_{\text{nl}} = 0.57\text{--}0.66$  on every dataset, while the exactly redundant reference and the noise column stay small. Because the random-forest residualization is not perfect — on Lipophilicity even the two null references have intervals that exclude zero — we drew a null distribution for  $\rho_{\text{nl}}$  with the same procedure: 500 random BID indices  $Mc$ ,  $c \sim N(0, I)$ , which are exactly in the linear span of the baseline (the relevant null for a redundant index), and 100 pure-noise columns, per dataset. The 99th percentile of  $|\rho_{\text{nl}}|$  under the in-span null is 0.11 (ESOL), 0.11 (FreeSolv), 0.08 (Lipophilicity) and 0.14 (BBBP); under pure noise it is 0.08, 0.10, 0.05 and 0.05. The null width is therefore sampling variance ( $\approx 2.6/\sqrt{n}$ ) plus, on ESOL, Lipophilicity and BBBP, an additional contribution from columns that lie in the span but that a forest residualizes imperfectly; on FreeSolv the two nulls coincide. Any residual association must be judged against this null width, not against zero (Figure 1b). Two candidate-dataset pairs exceed every one of the 500 in-span draws on their dataset: the degree entropy InfoH\_deg on ESOL ( $\rho_{\text{nl}} = 0.14$ ,  $[0.08, 0.20]$ ;  $p = 0.002$ ) and the four-cycle count on BBBP ( $-0.17$ ,  $[-0.21, -0.13]$ ;  $p = 0.002$ ); a third, InfoH\_deg on FreeSolv ( $-0.10$ ,  $[-0.18, -0.03]$ ), is exceeded by nine draws ( $p = 0.018$ ). After Benjamini–Hochberg adjustment over the 80 pairs the first two reach  $q = 0.08$  and the third  $q = 0.48$ ; 0 pairs are significant at  $q < 0.05$ . The two strongest also have linear intervals that exclude zero, so they are the only pairs with support from both residualizations; we call them suggestive. (The empirical  $p$ -values use the 100-tree statistic of the null run; Table 5 reports the 300-tree values, which differ by at most 0.004.) Several linear “signals” vanish once the baseline is allowed to act non-linearly — most strikingly the BBBP triangle count ( $0.225 \rightarrow 0.013$ ) — which means that their linear residual association was information the baseline already carries in non-linear form. Permutation  $p$ -values of a distance correlation between the residuals were not calibrated (the exactly redundant reference was rejected on two datasets) and are not used.

**Against a chemistry-aware baseline.** A QSPR practitioner’s baseline is not thirty topological indices but a standard descriptor set. We therefore repeated the linear screen with the baseline extended by RDKit 2D descriptors (the 180 of Section 8, of which 149–162 remain after removing non-finite, constant and collinear columns; none of the 30 topological indices is collinear with them) (Table 6). Against the union baseline 0 of the 20 candidates passes on any dataset. The two candidates with the largest statistics above become negligible; on Lipophilicity and BBBP most candidates are negligible, whereas on ESOL and FreeSolv most remain unresolved at these sample sizes (interval half-widths of about 0.06 and 0.09 with a baseline of rank about 180), so “none passes” is the accurate summary, not “all are negligible”. The reverse question has almost the same answer: of the 30 topological indices, 1 (the Mostar index on ESOL) retains an interval for  $|\text{pcor}|$  above 0.10 given the RDKit block, whereas 18–72 RDKit descriptors per dataset retain such signal given the topological block. Out of sample, adding all 30 indices and all 20 candidates to the RDKit descriptors improves a ridge model by at most 0.024 RMSE (ESOL; 3% of the reference, a point estimate from a single 5-fold split without an interval). Topology-only indices, new or old, add little to a chemistry-aware baseline on these targets.

**De-duplication.** Repeating the linear screen on the curated datasets of Table 1 changes  $|\text{pcor}|$  by at most 0.009 and changes no verdict.

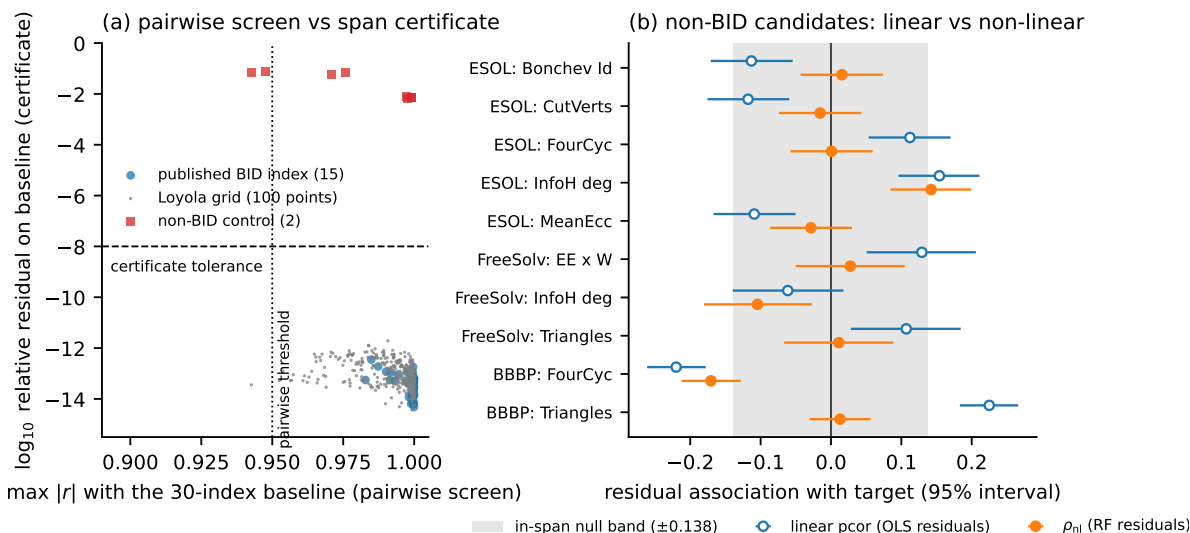


Figure 1: (a) The conventional pairwise screen (largest  $|r|$  with any of the 30 baseline indices) cannot separate the 15 published BID instances and the 100 Loyola points (all exactly in span, residual  $10^{-15}$ – $10^{-12}$ ) from the two non-BID controls (residual  $10^{-2}$ – $10^{-1}$ ): all lie at  $|r| \geq 0.94$ . The certificate separates them by ten orders of magnitude. (b) The candidate–dataset pairs of Table 5: linear partial correlation (hollow) and non-linear  $\rho_{nl}$  (filled) with 95% intervals; the shaded band is the widest of the four in-span null widths (99th percentile of  $|\rho_{nl}|$  over 500 exactly redundant draws; per-dataset values in the text). ESOL/InfoH\_deg and BBBP/FourCyc exceed every draw on their own dataset ( $p = 0.002$ ,  $q = 0.08$  over the 80 pairs); no pair reaches  $q < 0.05$ .

## 8 The information ceiling of topology-only descriptors

Every descriptor considered in this paper — all 30 baseline indices, all BID indices, and all 20 alternative candidates — is a function of the hydrogen-suppressed, atom- and bond-unlabelled molecular graph. Molecules with the same graph therefore receive identical descriptor vectors, whatever their atoms, bonds or stereochemistry. We grouped the curated molecules by a Weisfeiler–Lehman hash [51] of this graph (a canonical label built by four rounds of neighbourhood-label aggregation, identical for isomorphic graphs), and confirmed with an exact isomorphism test that every one of the 841 groups of molecules sharing a hash consists of isomorphic graphs; the grouping is therefore exact.

The collisions are extensive (Table 1): 55.5% of the curated ESOL molecules and 74.3% of the FreeSolv molecules share their graph with at least one other molecule; FreeSolv’s 639 molecules have only 236 distinct graphs, and the largest group contains 24 molecules. Within groups of molecules that share a graph, the pooled standard deviation of the target is 1.29 (ESOL), 3.18 (FreeSolv) and 0.62 (Lipophilicity), against overall standard deviations of 2.09, 3.85 and 1.20; on BBBP, 15.6% of the shared groups contain both labels. Because no topology-only model can separate molecules within a group, this within-group variance cannot be reduced by any such model (it also contains experimental noise, which limits every model). Note that the pooled within-group standard deviation is not the oracle’s error: molecules with a unique graph contribute zero within-group variance, which is why the oracle RMSE below is smaller. The sharpest statement uses the *graph oracle*: the in-sample predictor that assigns every molecule the mean target of its isomorphism class. No function of the unlabelled graph can beat it on the same data. Its  $R^2$  is 0.83 (ESOL), 0.57 (FreeSolv) and 0.95 (Lipophilicity), with RMSE 0.85, 2.53 and 0.26 (Figure 2); it is an in-sample bound for a dataset, not a population quantity,

Table 6: The target-aware screen against a chemistry-aware baseline.  $p_{\text{RD}}$ : RDKit 2D descriptors retained (finite, non-constant, not collinear);  $p_{\text{both}}$ : columns of the union with the 30 topological indices after removing exact collinearity (none of the 30 was collinear with the RDKit block).  $R^2$ : in-sample OLS. CV: 5-fold cross-validated RMSE (ridge, penalty by inner CV) or ROC-AUC (BBBP,  $\ell_2$ -logistic). Candidates: interval verdicts given the union baseline (pass / inconclusive / negligible; the remaining candidate is in span). Last column: number of the 30 topological indices whose interval for  $|\text{pcor}|$  given the RDKit block lies above 0.10.

| Dataset       | $p_{\text{RD}}$ | $p_{\text{both}}$ | in-sample $R^2$ |         |              | CV RMSE / AUC |         |              | candidates<br>pass/inc./negl. | topo.<br>beyond RD |
|---------------|-----------------|-------------------|-----------------|---------|--------------|---------------|---------|--------------|-------------------------------|--------------------|
|               |                 |                   | RD              | RD+topo | RD+topo+cand | RD            | RD+topo | RD+topo+cand |                               |                    |
| ESOL          | 161             | 191               | 0.929           | 0.940   | 0.944        | 0.729         | 0.731   | 0.705        | 0 / 13 / 6                    | 1                  |
| FreeSolv      | 149             | 179               | 0.972           | 0.976   | 0.977        | 1.051         | 1.044   | 1.043        | 0 / 18 / 1                    | 0                  |
| Lipophilicity | 158             | 188               | 0.591           | 0.607   | 0.617        | 0.829         | 0.824   | 0.820        | 0 / 3 / 16                    | 0                  |
| BBBP          | 162             | 192               | 0.545           | 0.559   | 0.566        | 0.891         | 0.892   | 0.894        | 0 / 1 / 18                    | 0                  |

and it is loose where most molecules have a unique graph (Lipophilicity: 28% shared). The out-of-sample counterpart — predicting each molecule from the other members of its class, and by the global mean when it has none — reaches only RMSE 1.87, 3.87 and 1.11, which shows that most of the information in these benchmarks is *not* carried by graph identity alone. The oracle nevertheless settles the FreeSolv question: the best topology-only model (Table 7) explains far less than the oracle allows, and the chemistry-aware reference below explains more out of sample ( $R^2 \approx 0.89$ , computed as  $1 - \text{RMSE}^2 / \text{Var}(y)$  from Table 7) than any topology-only model could in sample (0.57); the same holds on ESOL (0.90 versus 0.83). The comparison is fair in the direction that matters: with folds grouped by graph, the cross-validated predictions of any topology-only model are constant within an isomorphism class, so its cross-validated error is at least the in-sample oracle’s. The weak performance of topological indices on FreeSolv is therefore a property of the representation, not of any particular index.

Table 7: Downstream benchmark on the curated data:  $5 \times 5$  repeated cross-validation with folds grouped by graph, all preprocessing and selection inside the training folds. Top: mean RMSE (ESOL, FreeSolv, Lipophilicity) or ROC-AUC (BBBP), with the mean number of active features in parentheses. Bottom: mean paired difference (model minus reference; lower RMSE and higher AUC are better) with Nadeau–Bengio corrected 90% intervals; “eq.” marks equivalence (TOST) within 5% of the reference RMSE or 5% of the reference AUC above 0.5. \*On Lipophilicity the combined screen kept no feature in 24 of 25 folds (intercept-only model).

| Model                               | ESOL                          | FreeSolv                    | Lipophilicity               | BBBP (AUC)              |
|-------------------------------------|-------------------------------|-----------------------------|-----------------------------|-------------------------|
| full 30, LASSO / $\ell_1$ -logistic | 1.535 (16.0)                  | 3.516 (11.6)                | 1.130 (13.9)                | 0.808 (27.1)            |
| pairwise-screened, LASSO            | 1.580 (7.2)                   | 3.515 (5.9)                 | 1.139 (6.9)                 | 0.791 (5.8)             |
| combined-screened, LASSO            | 1.590 (4.3)                   | 3.519 (3.1)                 | 1.204 (0.0)*                | 0.791 (4.0)             |
| full 30, random forest              | 1.479 (30.0)                  | 3.443 (30.0)                | 1.086 (30.0)                | 0.831 (30.0)            |
| pairwise-screened, random forest    | 1.474 (7.5)                   | 3.429 (7.8)                 | 1.079 (8.0)                 | 0.847 (6.0)             |
| ten counts $m_{ij}$ , random forest | 1.458 (10.0)                  | 3.499 (10.0)                | 1.150 (10.0)                | 0.841 (10.0)            |
| ten counts $m_{ij}$ , LASSO         | 1.621 (8.8)                   | 3.498 (7.0)                 | 1.135 (8.6)                 | 0.800 (9.0)             |
| RDKit 2D descriptors, random forest | 0.674 (159.7)                 | 1.267 (148.9)               | 0.714 (160.2)               | 0.935 (172.5)           |
| Comparison                          | difference [corrected 90% CI] |                             |                             |                         |
| pairwise vs full (LASSO)            | +0.045 [+0.023, +0.066] eq.   | −0.001 [−0.072, +0.071] eq. | +0.009 [+0.004, +0.013] eq. | −0.018 [−0.036, +0.001] |
| combined vs full (LASSO)            | +0.055 [+0.031, +0.079]       | +0.004 [−0.078, +0.085] eq. | +0.073 [+0.060, +0.086]     | −0.017 [−0.035, +0.001] |
| pairwise vs full (RF)               | −0.006 [−0.032, +0.021] eq.   | −0.014 [−0.099, +0.071] eq. | −0.007 [−0.015, +0.000] eq. | +0.016 [+0.004, +0.028] |
| $m_{ij}$ vs full 30 (RF)            | −0.021 [−0.088, +0.046]       | +0.056 [−0.187, +0.299]     | +0.064 [+0.039, +0.089]     | +0.010 [−0.005, +0.025] |
| RDKit 2D vs full 30 (RF)            | −0.805 [−0.892, −0.718]       | −2.176 [−2.441, −1.911]     | −0.372 [−0.390, −0.354]     | +0.105 [+0.084, +0.126] |

**Benchmark.** To place the indices against this ceiling we ran a downstream benchmark on the curated data (Table 7). Folds are grouped by graph so that identical descriptor vectors never appear on both sides of a split; the design is  $5 \times 5$  repeated cross-validation; standardisation, screening, and the choice of the LASSO penalty all happen inside the training folds; and paired differences are tested with the Nadeau–Bengio variance correction [52], with equiv-

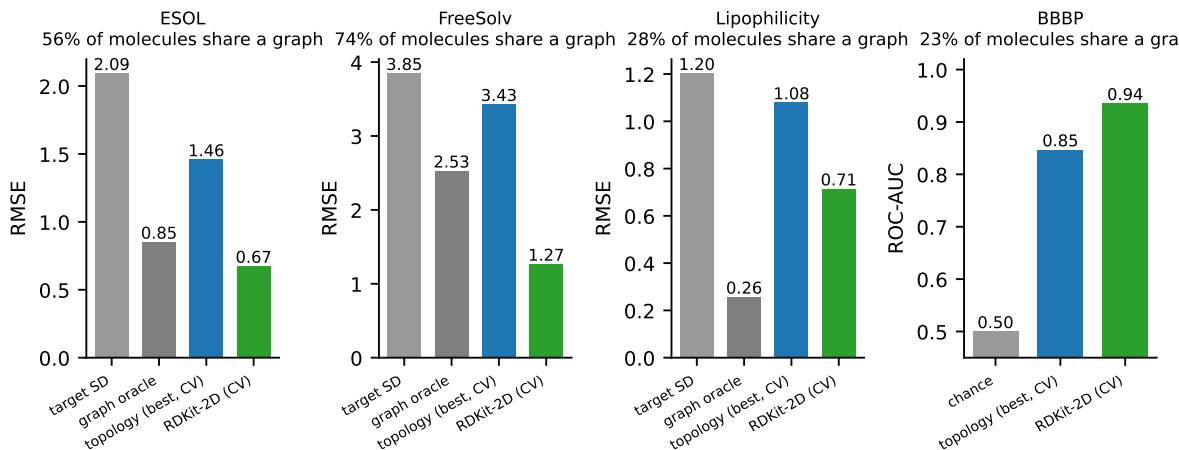


Figure 2: The information ceiling. Per dataset: the target standard deviation, the in-sample graph oracle (mean target of the isomorphism class; omitted for BBBP, where it is trivially near one), the best topology-only model of Table 7 (cross-validated), and the RDKit 2D random forest (cross-validated). On ESOL and FreeSolv the chemistry-aware model beats, out of sample, what any topology-only model could reach in sample.

alence assessed by two one-sided tests (TOST) [53] against a margin of 5% of the reference RMSE, or 0.05 ( $AUC_{\text{ref}} - 0.5$ ) for BBBP; the two one-sided tests at level 0.05 correspond to the reported 90% intervals. As a chemistry-aware reference we used a random forest on RDKit two-dimensional descriptors (180 of RDKit’s 208).<sup>1</sup>

Four results stand out. (i) The chemistry-aware reference is far ahead of every topology-only model — by 0.81 RMSE on ESOL, 2.18 on FreeSolv, 0.37 on Lipophilicity and 0.10 AUC on BBBP — and on ESOL and FreeSolv it exceeds, out of sample, the in-sample graph oracle of the topology-only representation. (ii) In a random forest the ten counts  $m_{ij}$  alone are not distinguishable from the full 30-index baseline on ESOL, FreeSolv and BBBP (intervals include zero, although on FreeSolv the interval is wide); only on Lipophilicity do the distance and spectral indices add measurably (+0.064 RMSE for the counts alone). (iii) Restricting the baseline to its pairwise-screened subset, and the combined screen, are discussed in Supplementary Section S4.

## 9 Discussion

### Practical guidance.

- *Proposing a new degree-based index:* report its weight vector  $(f(i, j))_{(i, j) \in P_4}$  and its coefficients on a classical basis (**certify**); do not report pairwise correlations as evidence of novelty; justify the index by parsimony, interpretation or mathematics, and compare its single-index  $R^2$  with ordinary least squares on the ten counts.
- *Modelling with topological descriptors:* use the ten counts  $m_{ij}$  directly instead of many BID indices; expect topology-only descriptors to add little to a chemistry-aware descriptor set ( $\leq 0.024$  RMSE here).
- *When the result does not apply:* indices outside the BID class (Remark 4.4), datasets with atoms of degree above four (Section 4.1), and non-linear models, for which only the

<sup>1</sup>The remaining 28 (EState- and VSA-EState-type descriptors, BalabanJ, BertzCT and Ipc) terminate the process under the RDKit 2022.09 / NumPy 2 combination used here and were excluded. Three of them are topological; their absence can only weaken the reference, which nevertheless leads by 0.37–2.18 RMSE.

empirical screen of Section 7 speaks.

**A concrete benchmark for BID indices.** Because every BID index lies in  $\text{col}[1, M]$ , its in-sample  $R^2$  in a single-index linear model can never exceed the  $R^2$  of ordinary least squares on the ten counts  $m_{ij}$ . On our data that bound is 0.42 (ESOL), 0.19 (FreeSolv), 0.12 (Lipophilicity) and 0.26 (BBBP, as a linear probability model), while the best of the 18 baseline indices, the 15 published instances and the 100 Loyola points reaches 0.36, 0.13, 0.05 and 0.16 (`results/bid_r2_bound.csv`). A proposer of a new BID index can therefore report, on any dataset, how close the index comes to this bound; a modeller who wants the degree information should simply use the ten counts.

**What a new degree-based index must show.** On molecular graphs (maximum degree four) a new BID index cannot be linearly independent of a baseline that includes ten generic classical degree-based indices; this is a theorem, not an empirical finding. Its justification must therefore come from elsewhere: parsimony and interpretability (one number that summarises the ten counts for a given property), mathematical properties, or behaviour under models that are not linear in the descriptors. We suggest that a proposal of a new BID index report its coefficient vector  $(f(i, j))_{(i, j) \in P_4}$ , which places it exactly in the common ten-dimensional space and makes coincidences between differently named indices immediately visible.

**What the target-aware screen adds.** For indices outside the BID class the question is genuinely empirical, and the answer depends on the baseline and on uncertainty. Against a chemistry-aware baseline none passes (Table 6); against the topological baseline of the point-estimate “passers” in an earlier version of this analysis, none survives on ESOL or FreeSolv once intervals are reported, and most linear signals do not survive non-linear residualization. A pairwise threshold, which ignores both the multivariate baseline and the target, is not a meaningful novelty criterion.

**What topology cannot do.** Section 8 shows that a large share of molecules in these benchmarks are indistinguishable to every hydrogen-suppressed, unlabelled-graph descriptor. This is not a defect of any particular index; it bounds the whole class and explains the weak predictive performance of topological indices on targets such as FreeSolv.

## 10 Limitations

- The certificate concerns linear redundancy; it does not rank BID indices by usefulness, and it is silent on non-linear models except through Section 7.
- The target-aware analyses use four MoleculeNet benchmarks. The certificate itself was checked on thirteen; datasets with hypervalent or metal centres (HIV) leave  $\mathcal{G}_4$  and require either a larger baseline or the counts  $m_{ij}$  themselves, which the data-level rank check of Proposition 4.2 detects.
- The graph oracle is an in-sample bound for a given dataset; the RDKit reference omits 28 descriptors for software reasons (footnote in Section 8).
- The effect-size margin  $\tau = 0.10$  is a convention; we report intervals so that readers can apply their own.
- The non-linear check relies on random-forest residualization, whose null distribution for exactly redundant columns has a 99th percentile of 0.08–0.14 depending on the dataset; effects below that width are not resolved, and with 500 draws per dataset the smallest

attainable  $p$ -value is 0.002, so a single pair can reach at best  $q \approx 0.08$  after correction over 80 pairs.

- BBBP partial correlations use a linear probability model.
- The candidate set of Section 7 is illustrative, not exhaustive.

## 11 Conclusion

We showed that, for bond-incident-degree indices on molecular graphs, redundancy is not a matter of correlation thresholds but of linear algebra: once a baseline of classical indices spans the edge-type counts, every such index — published or yet to be proposed — is an exact linear combination of the baseline and can contribute no linear information about any target. We verified this certificate across the MoleculeNet benchmarks and on a census of published degree-based indices, diagnosed the one dataset on which it fails, and showed how statistical diagnostics computed without this knowledge manufacture spurious signal. For indices outside the class we replaced the pairwise screen by a target-aware one with intervals, a calibrated non-linear check and a chemistry-aware baseline, and found that the little residual signal that survives against a topological baseline disappears against a chemistry-aware one. Finally, we quantified the information ceiling imposed by graph collisions, which bounds every topology-only descriptor and explains why chemistry-aware descriptors outperform them.

## Data and code availability

Code, curated-data scripts, results and figures are available at <https://github.com/Singati2/topological-index-orthogonality>; the snapshot reported here is git tag v2.1-paper2. Datasets are downloaded from the MoleculeNet distribution by `src/load_data.py`. The analyses of this paper are produced by the following scripts in `scripts/`: 70 (certificate, census, identities), 02, 16 and 17 (screens), 73 (non-linear check), 74 (de-duplication), 72 (curation, collisions, benchmark), 76 (dataset verification), 77 (exact isomorphism check,  $R^2$  bound), 78 (certificate on thirteen datasets), 79 (screen against the RDKit baseline), 80 (null distribution of  $\rho_{nl}$ ), 81 (figures) and 75 (tables and the numerical macros used in the text); the one-function certificate is `src/certify.py`. The four datasets are included in `data/`, verified byte-identical to the MoleculeNet distribution. All analyses used Python 3.11, RDKit 2022.09, scikit-learn 1.8 and NumPy 2.4 with the seeds recorded in each script (20261005 unless stated).

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## Supplementary Information

### S1 The baseline

Table S1: The 30-index classical baseline.

| Family                 | Indices                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Degree-based, BID (18) | $M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA$ [47], $AG, ABC, ABS, AZI, SO$ [5], $SO_{\text{red}}, Alb$ [48], $\sigma, \text{red}M_1$ |
| Distance-based (8)     | $W, WW, HR, \text{degree distance}, \text{Gutman}, \text{Mostar}, Sz, PI$                                                               |
| Spectral (3)           | Estrada EE, energy, spectral radius                                                                                                     |
| Other (1)              | Balaban $J$                                                                                                                             |

### S2 The value matrix of the baseline

Table S2: The value matrix  $\Phi_B$  of the 18 degree-based baseline indices on the ten degree pairs  $(i, j) \in P_4$ : entry  $f_b(i, j)$ , obtained by evaluating each index on  $K_{i,j}$  and dividing by its  $ij$  edges. Vertex-degree sums  $\sum_v g(d_v)$  are written as  $f(a, b) = g(a)/a + g(b)/b$ . The matrix has rank 10 (smallest/largest singular value  $1.2e - 04$ ).

| Index             | $f(a, b)$                          | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (2, 2) | (2, 3) | (2, 4) | (3, 3) | (3, 4) | (4, 4) |
|-------------------|------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $M_1$             | $a + b$                            | 2      | 3      | 4      | 5      | 4      | 5      | 6      | 6      | 7      | 8      |
| $M_2$             | $ab$                               | 1      | 2      | 3      | 4      | 4      | 6      | 8      | 9      | 12     | 16     |
| $mM_1$            | $a^{-3} + b^{-3}$                  | 2      | 1.12   | 1.04   | 1.02   | 0.25   | 0.162  | 0.141  | 0.0741 | 0.0527 | 0.0312 |
| $mM_2$            | $1/(ab)$                           | 1      | 0.5    | 0.333  | 0.25   | 0.25   | 0.167  | 0.125  | 0.111  | 0.0833 | 0.0625 |
| $F$               | $a^2 + b^2$                        | 2      | 5      | 10     | 17     | 8      | 13     | 20     | 18     | 25     | 32     |
| $R$               | $(ab)^{-1/2}$                      | 1      | 0.707  | 0.577  | 0.5    | 0.5    | 0.408  | 0.354  | 0.333  | 0.289  | 0.25   |
| $\text{SCI}$      | $(a + b)^{-1/2}$                   | 0.707  | 0.577  | 0.5    | 0.447  | 0.5    | 0.447  | 0.408  | 0.408  | 0.378  | 0.354  |
| $H$               | $2/(a + b)$                        | 1      | 0.667  | 0.5    | 0.4    | 0.5    | 0.4    | 0.333  | 0.333  | 0.286  | 0.25   |
| $GA$              | $2\sqrt{ab}/(a + b)$               | 1      | 0.943  | 0.866  | 0.8    | 1      | 0.98   | 0.943  | 1      | 0.99   | 1      |
| $AG$              | $(a + b)/(2\sqrt{ab})$             | 1      | 1.06   | 1.15   | 1.25   | 1      | 1.02   | 1.06   | 1      | 1.01   | 1      |
| $ABC$             | $\sqrt{(a + b - 2)/(ab)}$          | 0      | 0.707  | 0.816  | 0.866  | 0.707  | 0.707  | 0.707  | 0.667  | 0.645  | 0.612  |
| $ABS$             | $\sqrt{(a + b - 2)/(a + b)}$       | 0      | 0.577  | 0.707  | 0.775  | 0.707  | 0.775  | 0.816  | 0.816  | 0.845  | 0.866  |
| $AZI$             | $(ab/(a + b - 2))^3$ (0 on $K_2$ ) | 0      | 8      | 3.38   | 2.37   | 8      | 8      | 8      | 11.4   | 13.8   | 19     |
| $SO$              | $\sqrt{a^2 + b^2}$                 | 1.41   | 2.24   | 3.16   | 4.12   | 2.83   | 3.61   | 4.47   | 4.24   | 5      | 5.66   |
| $SO_{\text{red}}$ | $\sqrt{(a - 1)^2 + (b - 1)^2}$     | 0      | 1      | 2      | 3      | 1.41   | 2.24   | 3.16   | 2.83   | 3.61   | 4.24   |
| $Alb$             | $ a - b $                          | 0      | 1      | 2      | 3      | 0      | 1      | 2      | 0      | 1      | 0      |
| $\sigma$          | $(a - b)^2$                        | 0      | 1      | 4      | 9      | 0      | 1      | 4      | 0      | 1      | 0      |
| $\text{red}M_1$   | $(a - 1)^2/a + (b - 1)^2/b$        | 0      | 0.5    | 1.33   | 2.25   | 1      | 1.83   | 2.75   | 2.67   | 3.58   | 4.5    |

### S3 Definitions of the 20 alternative candidate indices

For reproducibility we give the formal definition of each of the twenty candidate indices screened in Section 7. Let  $G = (V, E)$  be a simple connected hydrogen-suppressed molecular graph with  $n = |V|$  vertices,  $m = |E|$  edges, degree sequence  $\{d_v\}_{v \in V}$ , geodesic distance matrix  $D = (d_{uv})$ , adjacency matrix  $A$ , eccentricity  $\varepsilon(v) = \max_u d_{uv}$ , and adjacency-spectrum / Laplacian-spectrum eigenvalues  $\{\lambda_i\} / \{\mu_i\}$  respectively. All twenty definitions are implemented in `src/novel_candidates.py` of the companion repository.

**Information-theoretic (5).** With  $H(\{c_i\}) = -\sum_i (c_i / \sum_j c_j) \log(c_i / \sum_j c_j)$  the Shannon entropy of a count distribution:

$$\begin{aligned}\text{InfoH\_deg}(G) &= H(\{|\{v : d_v = k\}|\}_k), \\ \text{InfoH\_dist}(G) &= H(\{|\{(u, v) : d_{uv} = k, u < v\}|\}_k), \\ \text{InfoH\_spec}(G) &= -\sum_i p_i \log p_i, \quad p_i = \lambda_i^2 / \sum_j \lambda_j^2, \\ \text{InfoH\_ecc}(G) &= H(\{|\{v : \varepsilon(v) = k\}|\}_k), \\ \text{Bonchev\_Id}(G) &= W \log_2 W - \sum_{k \geq 1} k d_k \log_2(k d_k),\end{aligned}$$

where in the last line  $d_k = |\{(u, v) : d_{uv} = k, u < v\}|$  and  $W = \sum_k k d_k$  is the Wiener index; this Bonchev-type form, as implemented, differs from the standard information-theoretic Wiener index  $I_D^W = W \log_2 W - \sum_k k d_k \log_2 k$ .

**Centrality-based (5).** With NetworkX centralities (betweenness normalised by  $\binom{n-1}{2}$ , NetworkX's default):

$$\begin{aligned}\text{Btw\_sum}(G) &= \sum_{v \in V} c_B(v), \quad c_B(v) = \binom{n-1}{2}^{-1} \sum_{s \neq t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}}, \\ \text{Cls\_sum}(G) &= \sum_v (n-1) \left( \sum_{u \neq v} d_{uv} \right)^{-1}, \\ \text{Eig\_sum}(G) &= \sum_v x_v, \quad x \text{ the (positive) Perron eigenvector of } A, \\ \text{Harm\_cent}(G) &= \sum_v \sum_{u \neq v} \frac{1}{d_{uv}} = 2 \text{HR}(G), \\ \text{Btw\_var}(G) &= \text{Var}_v(c_B(v)).\end{aligned}$$

**Spectral-hybrid (3).** With  $\text{diam}(G) = \max_{u,v} d_{uv}$  and Estrada index  $\text{EE}(G) = \sum_i e^{\lambda_i}$ :

$$\begin{aligned}\text{AlgConn}(G) &= \mu_2(G), \text{ second-smallest Laplacian eigenvalue,} \\ \text{Rho} \times \text{Dia}(G) &= \rho(G) \cdot \text{diam}(G), \quad \rho(G) = \max_i |\lambda_i|, \\ \text{EE} \times \text{W}(G) &= \log(1 + \text{EE}(G)) \cdot \log(1 + W(G)).\end{aligned}$$

**Motif / structural (4).**

$$\begin{aligned}\text{Triangles}(G) &= \frac{1}{6} \text{tr}(A^3), \\ \text{MeanClust}(G) &= \frac{1}{n} \sum_v \frac{2t(v)}{d_v(d_v - 1)}, \quad t(v) = \text{triangles at } v, \\ \text{FourCyc}(G) &= \frac{1}{8} (\text{tr}(A^4) - 2 \sum_v d_v(d_v - 1) - 2m), \\ \text{CutVerts}(G) &= |\{v \in V : G - v \text{ is disconnected}\}|.\end{aligned}$$

**Eccentricity-based (3).**

$$\begin{aligned}\text{TotEcc}(G) &= \sum_{v \in V} \varepsilon(v), \\ \text{Radius}(G) &= \min_{v \in V} \varepsilon(v), \\ \text{MeanEcc}(G) &= \frac{1}{n} \sum_{v \in V} \varepsilon(v).\end{aligned}$$

None of these twenty candidates is claimed as a newly invented index. The information-theoretic family follows the information-theoretic descriptor tradition [?]; the centrality family follows the standard graph-theory literature; the motif and eccentricity families are textbook graph invariants; the spectral-hybrid combinations are deliberately constructed novel pairings to test whether non-standard product indices escape the redundancy ceiling. They appear here purely as screening targets to test the generality of the target-aware screen, and the verdict reported in Section 7 is a statement about this particular candidate set under the tested baselines and thresholds, not a claim that any individual candidate is or is not a useful QSPR descriptor.

## S4 Statistical details of the benchmark

The benchmark of the main text uses  $r = 5$  repeats of  $k = 5$ -fold cross-validation with folds grouped by isomorphism class of the hydrogen-suppressed graph. For a paired difference  $d_{ij}$  (repeat  $i$ , fold  $j$ ) between two models, the Nadeau–Bengio corrected variance of the mean difference is  $\hat{\sigma}^2(1/(kr) + n_{\text{test}}/n_{\text{train}})$ , where  $\hat{\sigma}^2$  is the sample variance of the  $kr$  differences; the corrected  $t$  statistic has  $kr - 1$  degrees of freedom and the reported 90% intervals are  $\bar{d} \pm t_{0.95, kr-1} \hat{\sigma}_{\text{corr}}$ . Equivalence is tested by two one-sided tests (TOST) at level 0.05 against a margin of 5% of the reference model’s mean RMSE, or 0.05 ( $\text{AUC}_{\text{ref}} - 0.5$ ) for BBBP; equivalence is declared when the 90% interval lies inside  $[-\delta, \delta]$ . Within each dataset the five comparisons are Holm-adjusted. The inner cross-validation that selects the LASSO or  $\ell_1$ -logistic penalty is also grouped by graph. Random forests use 300 trees.

**Screened subsets and the combined screen.** Restricting the baseline to its pairwise-screened subset (about 6–8 indices) is equivalent within 5% for the random forest on the three regression sets and slightly better on BBBP; for LASSO it is equivalent within the margin on the regression sets, although the cost is distinguishable from zero on ESOL (+0.045) and Lipophilicity (+0.009), and it is not shown equivalent on BBBP. The combined screen, which also requires  $|\text{pcor}| \geq 0.10$ , is a diagnostic and not a feature selector: on Lipophilicity it removes every feature in 24 of 25 folds.

## S5 Datasets used for the generality check

The nine additional MoleculeNet files (BACE, ClinTox, SIDER, Tox21, ToxCast, HIV, MUV, QM8, QM9) are downloaded from the DeepChem mirror by script 78 of the repository; their SHA-256 digests are recorded in the results file `certificate_generality.csv`. Only the SMILES column of each file is used.